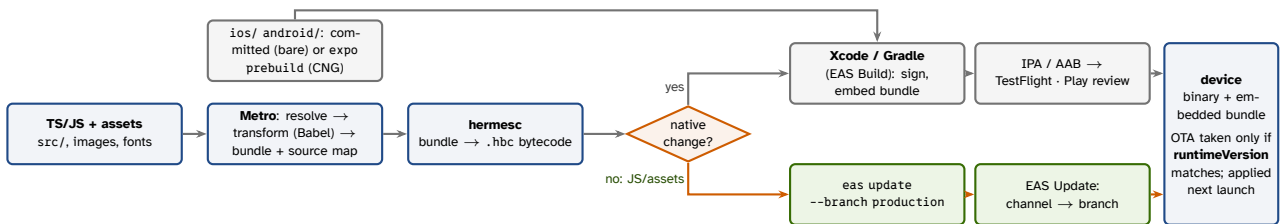


**In one line:** An RN app is a **native binary** (reviewed, signed, store-shipped) that **runs a JS bundle** made by Metro. Change native code  $\Rightarrow$  new store build; change only JS/assets  $\Rightarrow$  an **OTA update** may replace the bundle — but only for binaries with the same **runtimeVersion**.



per build AND per OTA update upload to the crash reporter: JS source map (Hermes: composed with the hermes map) · iOS dSYM · Android R8 mapping.txt — else prod stacks are unreadable  
 native change = new pod/Gradle dep, permission, Info.plist/entitlement, icon/splash, RN/Expo SDK bump, config-plugin change · otherwise the gate says OTA

## Expo vs bare — when each

- **CNG** (Continuous Native Generation): ios/ android/ are **build output**, regenerated by `npx expo prebuild --clean` from `app.json/app.config.ts` + **config plugins** (withInfoPlist, withEntitlementsPlist, withAndroidManifest). Native tweaks live in a plugin — a hand edit is lost on the next prebuild. Upgrades become “regenerate”.
- **Expo Go** = a prebuilt sandbox with one SDK’s modules: no custom native code. A **dev build** (`expo-dev-client`) is *your* binary + dev menu — needed once any native library is added.
- **Bare** (`npx @react-native-community/cli init`): you own the Xcode/Gradle projects. Choose it for **brownfield** (RN inside an existing native app) or when a native team owns the projects. Default for new apps: Expo (the RN docs recommend a framework). Expo  $\neq$  “no native code”.

## EAS · OTA updates

- **Build**: cloud builders, `eas.json` profiles, managed certificates / provisioning profiles / keystore (`--local` to build on your Mac). **Submit**: `eas submit -p ios`  $\rightarrow$  App Store Connect / TestFlight; Android  $\rightarrow$  a Play track.
- **Update**: a build carries a **channel**; a channel points at a **branch** of updates. Promote/roll back by re-pointing the channel or republishing an older update. `expo-updates` checks at launch, downloads, applies on the *next* cold start (`checkForUpdateAsync` / `fetchUpdateAsync` / `reloadAsync` for urgent fixes); a launch crash falls back to the previous update.
- **runtimeVersion** policies: `appVersion`, `nativeVersion`, `fingerprint` (hash of native inputs — safest), or a manual string.
- **Apple**: downloaded JS is allowed if it does not change the app’s **primary purpose**, add a storefront, or bypass security — fixes and tweaks, not a different app (*unverified*). Hosted **CodePush** died with App Center (retired 31 Mar 2025); Microsoft left a self-hostable code-push-server.

## Example — eas.json + app.json

```

{ "build": { // eas.json
  "development": { "developmentClient": true },
  "preview": { "channel": "preview" },
  "production": { "channel": "production" } } }
// app.json > expo:
"runtimeVersion": { "policy": "fingerprint" },
"updates": { "url": "https://u.expo.dev/<project-id>" }

```

Related: rn-native-modules (spec = ABI) · rn-navigation · Hermes · code signing · TestFlight · CI/CD

## Metro · env · upgrades · debugging

- **Metro** resolves `.ios.tsx/.android.tsx/.native.tsx` first, caches transforms (stale after a Babel/env change: `npx expo start -c --reset-cache`); monorepos need `watchFolders`. Debug loads from Metro (:8081).
- **Fast Refresh** re-runs the edited module, keeps component state; a file exporting non-components remounts.
- **Secrets**: the bundle is public. `EXPO_PUBLIC_*` / `react-native-config` values are *inlined*. Keys stay on a server.
- **Upgrades hurt**: the native template moves (AppDelegate, Podfile, AGP, Kotlin, min iOS), every native dep must support the new RN, Xcode bumps too. **Upgrade Helper** = template diff per version. Small steps; Expo: `npx expo install --fix + npx expo-doctor`.
- **Debug: React Native DevTools** (Chrome DevTools on Hermes) — default since 0.76; **Flipper** left the template in 0.74. Native: Xcode, Instruments.

## iOS · Android specifics

- **iOS**: Podfile `use_react_native! + use_native_modules!`; `bundle exec pod install`; open the **.xcworkspace**. Envs = `schemes/configurations` + `xcconfig` (own bundle ID). Build phase “Bundle React Native code and images” embeds `main.jsbundle` in Release.
- **Android**: `gradle.properties` (`hermesEnabled`; `newArchEnabled=false` is ignored since 0.82); envs = `productFlavors`; ship an **AAB** with Play App Signing; **R8** (`enableProguardInReleaseBuilds`) shrinks and obfuscates — reflection-using libs need keep rules.

## Traps

- ! OTA JS needing a new native module  $\rightarrow$  crash; use fingerprint.
- ! Hand-editing ios/ in a CNG project — gone after prebuild.
- ! API key in `.env` “hidden” — it is in the bundle.
- ! No source maps for an OTA update  $\rightarrow$  unreadable prod stacks.

## Remember

Native  $\Rightarrow$  store · JS  $\Rightarrow$  OTA · runtimeVersion is the contract.